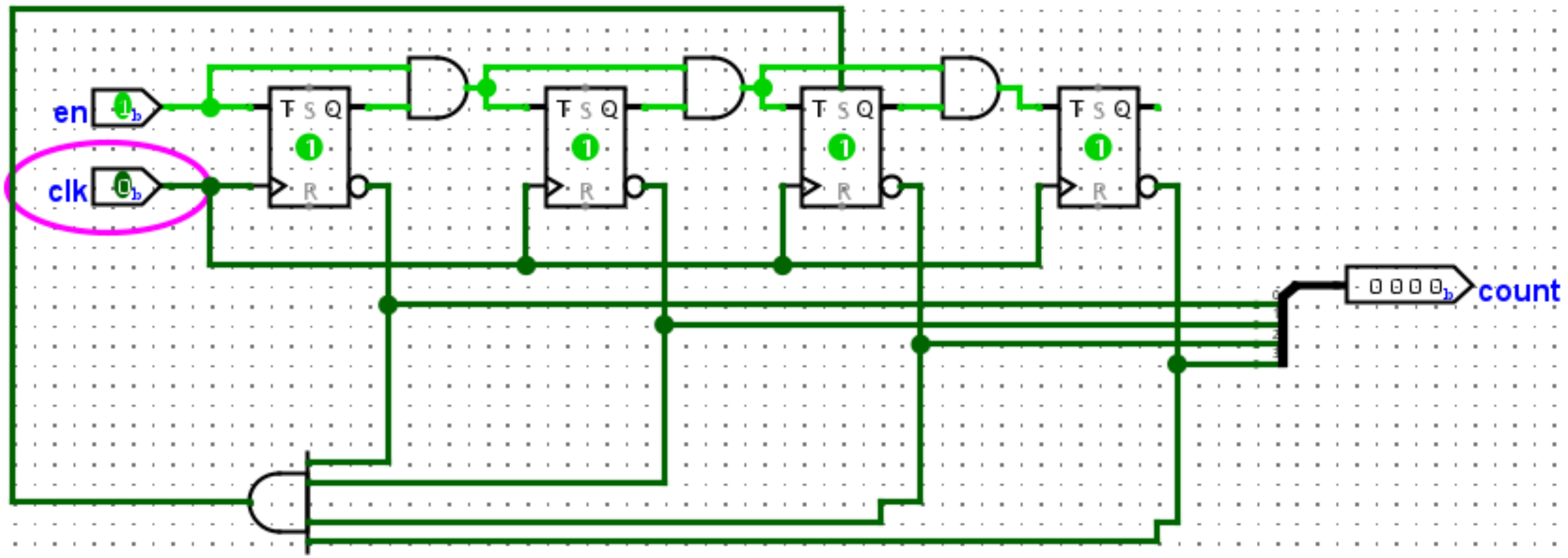
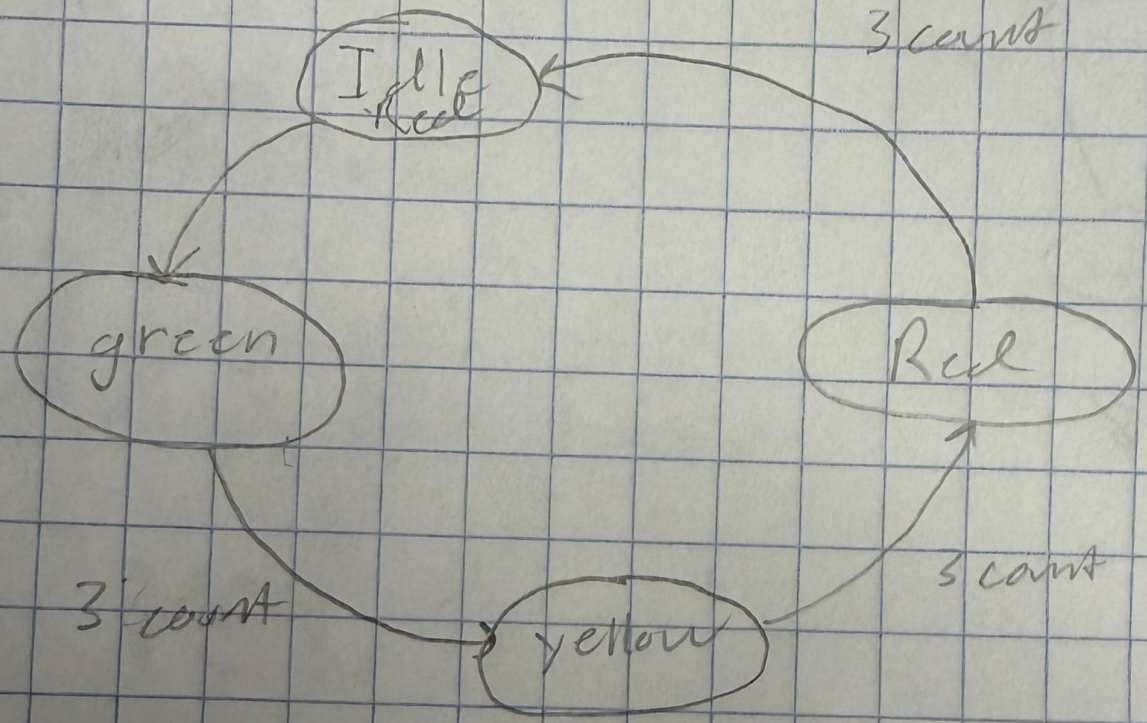
Synchronous Down Counter from 3 to 0 with reset

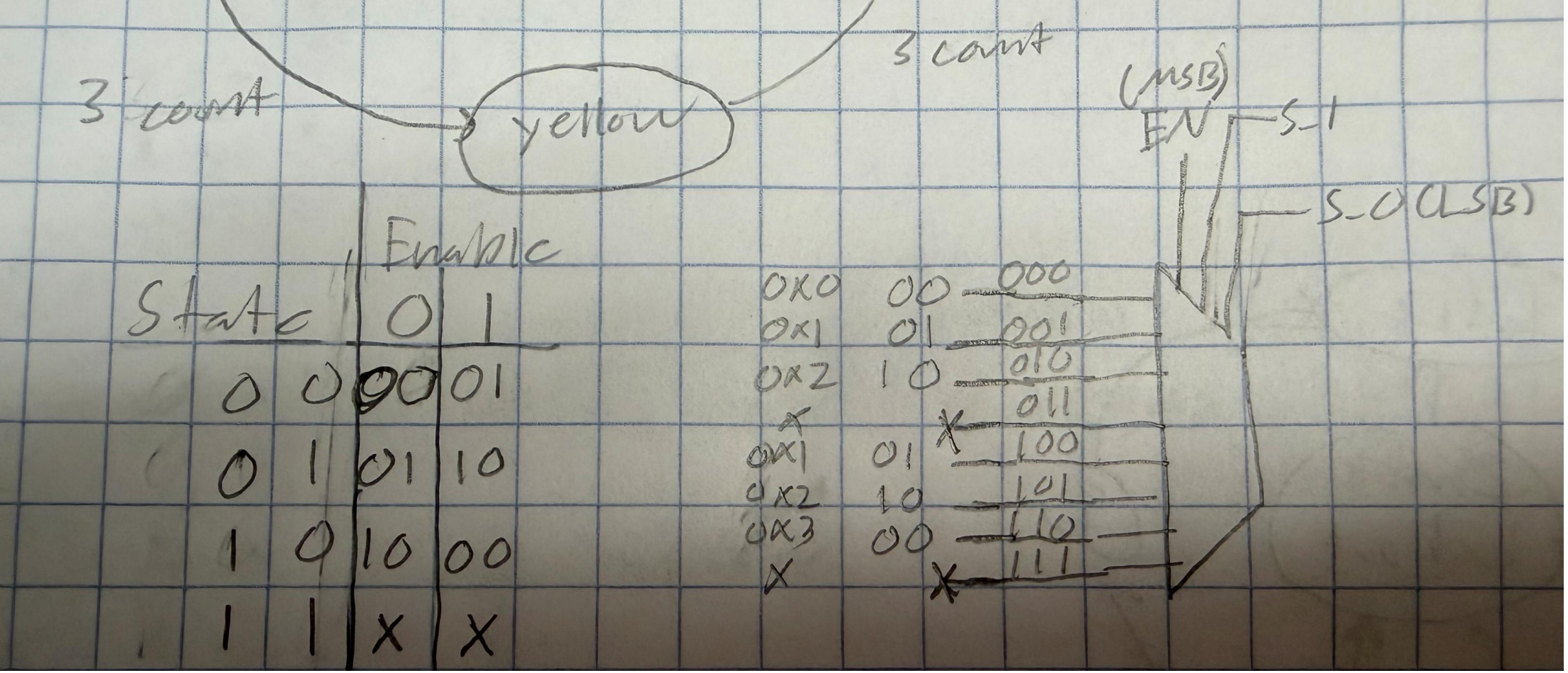


Digital Design
Nov 3, 2026

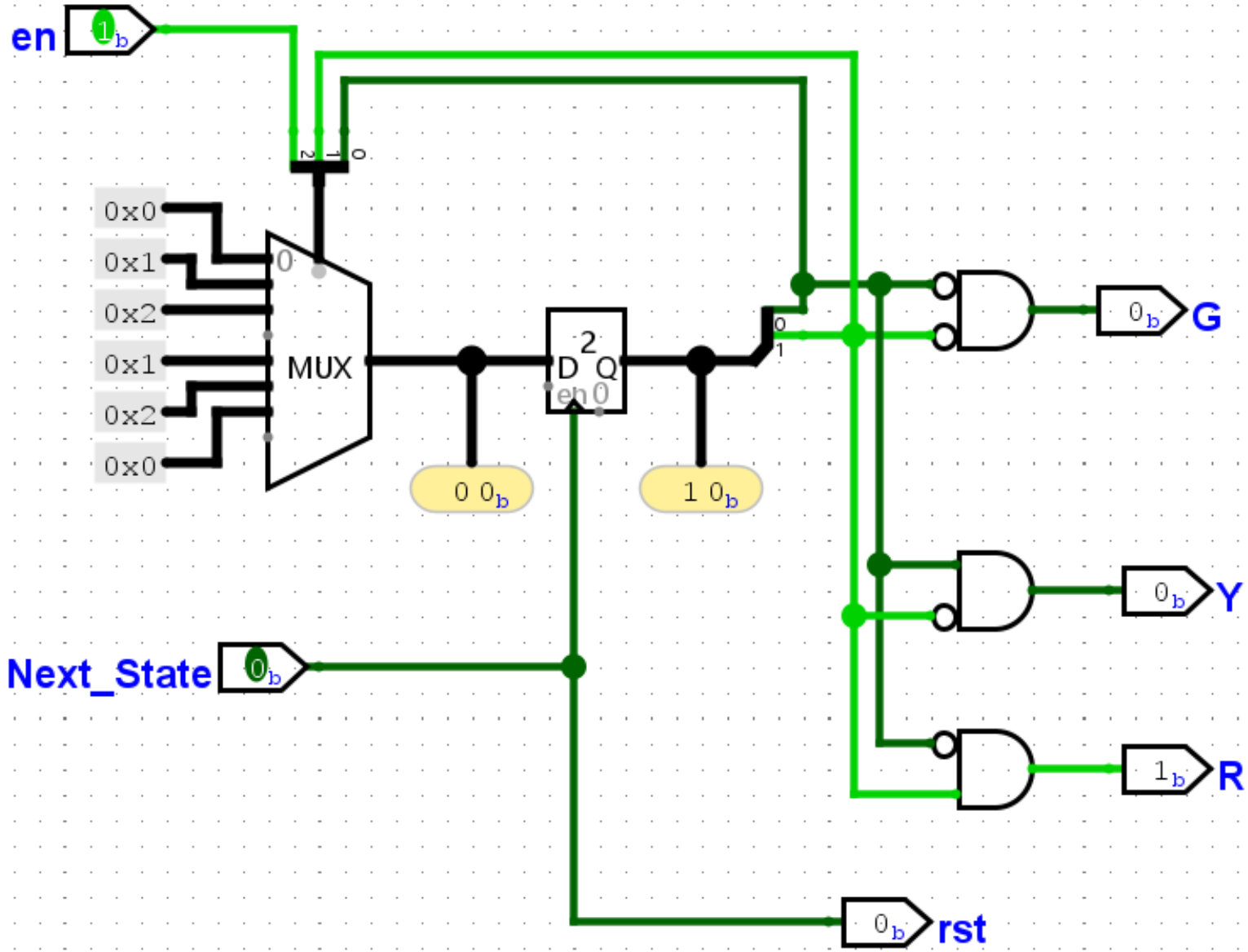
Lab #7 - State machine for traffic



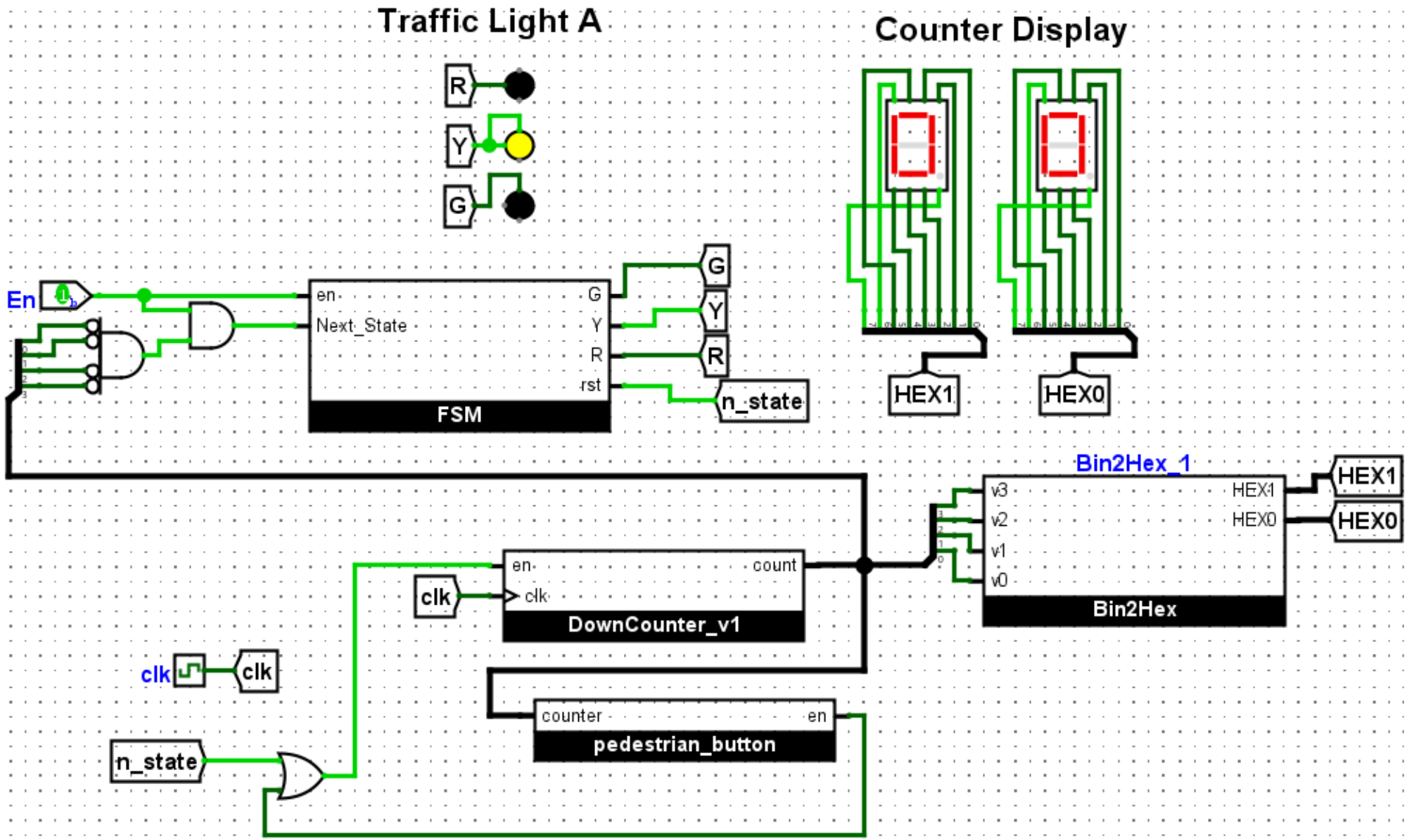
State Machine #1



Truth Table and MUX design



Accurately changed between the three states in the correct order


Traffic Light A
Counter Display
R
Y
G
En
en
Next_State
G
Y
R
rst
FSM
n_state
HEX1
HEX0
Bin2Hex_1
v3
v2
v1
v0
HEX1
HEX0
Bin2Hex
clk
en
clk
count
DownCounter_v1
clk
clk
counter
en
pedestrian_button
n_state